Why Students Order Out

Analysing Mess Food Habits and Outside Food Ordering Habits among Hostel Students

IDHV 2026M Project Report | Introduction to Data Handling and Visualization

Group number	[to be filled]
Group members	[names and roll numbers of all members]
Data collected	Online survey, 12 Oct 2026 to 16 Oct 2026; 55 respondents

Abstract

We surveyed 55 hostel students about two things: how they experience the mess, and how they order food from outside. Using descriptive statistics and visual exploratory analysis, we found that dinner is the least-liked mess meal (73% dissatisfied) and also the meal that 93% of students order outside food for. A typical student orders 2 times a week and spends about Rs 300, roughly Rs 150 per order, and weekly spending rises steadily with the number of orders. Ordering is mostly social (66% of orders are with friends), and it is rare on Wednesdays and Sundays, the days the mess serves non-veg meals. Skipping mess meals, disliking the menu and low mess ratings tend to go together with more ordering and higher spending, although these links are weaker than the link between the number of orders and spending.

1. Introduction

1.1 Motivation

Hostel students already pay for the mess, yet many of them regularly pay again for food from outside. For students this is a recurring expense, and for mess management it is a signal that something about the mess is not working. Understanding *when*, *how often*, *with whom* and *why* students order outside food is a first step to understanding where the mess falls short and how much money students spend as a result.

1.2 Research question

Main question: How do students' mess food habits and outside food ordering habits look on their own, and how are the two related?

- **Mess habits alone:** How satisfied are students with each mess meal, with taste and quality? How many meals do they skip, and how often do they dislike the menu?
- **Outside ordering habits alone:** On which days, how often and with whom do students order? How much do they spend? What reasons do they give?
- **Relationships:** Is the least-liked meal the one students replace? Do more orders mean more spending? Do skipping meals, a disliked menu or low satisfaction go together with more ordering?

1.3 Objectives

- Design a short questionnaire and collect responses from more than 50 students (quantitative and qualitative variables).
- Clean and organise the raw data: encode, handle outliers, derive new variables and normalise where needed.
- Describe each habit with simple summary statistics and charts, and uncover relationships with visual exploratory data analysis.

2. Methodology

2.1 Research design and sampling

We ran a cross-sectional online survey (Google Forms) among hostel students of IIITS who eat in the mess. The form was shared in student groups and collected 55 responses between 12 Oct 2026 and 16 Oct 2026. We used **convenience sampling** because it is quick and free and fits the scale of a course project. Its drawback is that respondents are not a random sample: 46 of 55 (84%) are third-year students, so the results describe this group rather than the whole hostel. Ordering and mess habits were asked about a *typical week* to keep answers simple.

Context used in the analysis: the mess serves non-veg meals (for example chicken curry and biryani) on Wednesdays and Sundays. This is the group's knowledge of the weekly menu; the survey did not ask about it directly.

2.2 Questionnaire and variables

The questionnaire has three parts: (i) outside ordering behaviour, (ii) mess satisfaction and habits, and (iii) year of study. It mixes single-choice, multiple-choice, 5-point rating and numeric questions. The full list of questions is in the Appendix.

Variable group	Variables	Type
Outside ordering (quantitative)	Weekly spend (Rs); orders per week; orders alone; orders with friends	Numeric (ratio / count)
Outside ordering (qualitative)	Usual ordering company; reasons (multi-select); meal ordered for; days ordered (multi-select); ordered outside mess hours; ordered after skipping a meal	Nominal / binary
Mess satisfaction	Taste (1-5); quality and consistency (1-5); satisfaction with breakfast, lunch, snacks, dinner (5-point scale)	Ordinal
Mess habits	Meals skipped per week; days menu disliked (of 7); how often the menu influences ordering (4 levels); favourite meal	Count / ordinal / nominal
Background	Year of study	Nominal

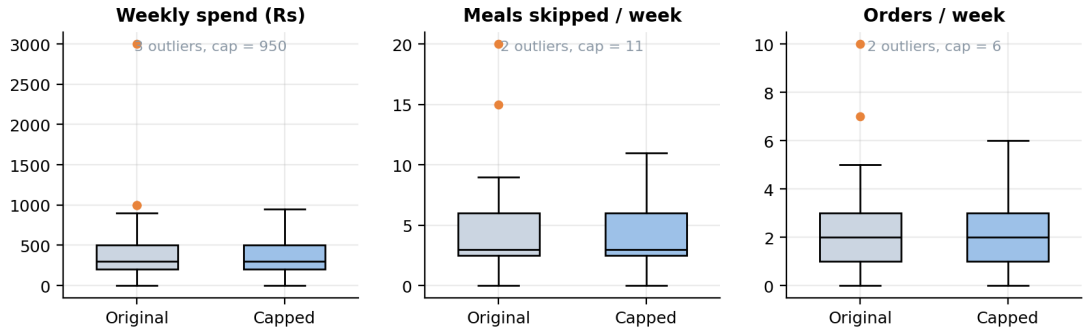
The survey therefore has 21 analysis variables and 55 respondents, which satisfies the project requirement of more than 50 data points and more than 3 variables with both quantitative and qualitative types.

3. Data Preprocessing

Step	What was done	Why
1. Inspection	55 rows and 22 columns checked. 0 missing values and 0 duplicate rows found.	No imputation or removal needed.
2. Tidying	Columns renamed to short names; timestamp dropped from the cleaned file; text trimmed.	Easier analysis.
3. Encoding	Satisfaction answers coded 1 (very dissatisfied) to 5 (very satisfied); menu influence coded 1-4; Yes/No coded 1/0.	Needed for averages and correlations.
4. Multi-select split	Days ordered split into 7 binary columns; reasons split into 9 binary columns.	A multi-select answer is not analysable as one text cell.
5. Validation	Orders alone + orders with friends equals total orders in 55 of 55 rows.	Confirms respondents answered consistently.
6. Derived variables	Spend per order; share of orders with friends; overall mess index (mean of taste, quality and four meal ratings); number of order days; ordered on a non-veg day; order, skip and mess-satisfaction groups.	Simple, interpretable measures. The six ratings are consistent with each other (Cronbach's alpha = 0.76).
7. Outliers	IQR rule (above Q3 + 1.5 x IQR). Weekly spend: 3 values above Rs 950 (1000, 1000, 3000). Meals skipped: 2 above 11.2 (15, 20). Orders: 2 above 6 (7, 10). Values were capped at the fence in new columns ending in _capped; original columns were kept.	With only 55 rows, deleting students would lose information. The Rs 3000 answer could be a typing error or a genuine heavy spender, so we limit its influence rather than drop it.
8. Normalisation	Min-max scaling (0 to 1) of capped spend, capped orders, capped skips, disliked days and the mess index.	Puts variables with different units on one scale so they can be compared in one chart (Figure 17).

Descriptive statistics in Section 4 use the **original** values; the scatter plots and the correlation heatmap use the **capped** values so that a few extreme answers do not dominate.

Figure 1. Outlier treatment (IQR rule): before and after capping



4. Results and Analysis

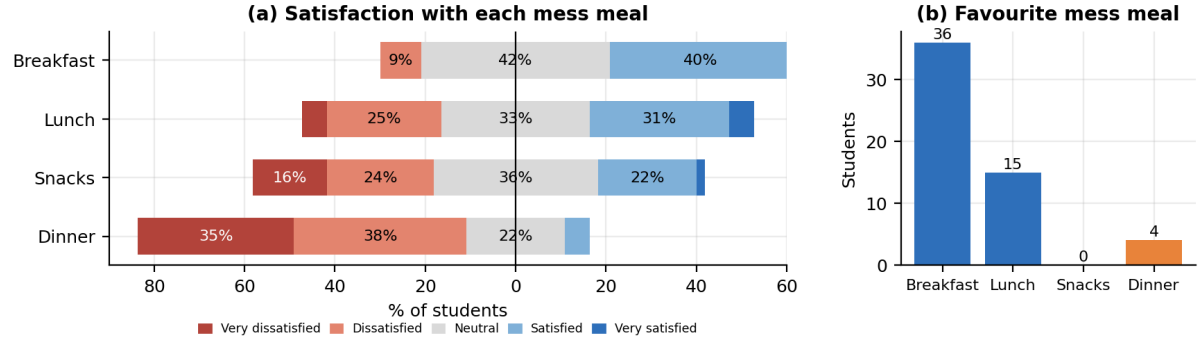
4.1 Summary statistics

Variable	Mean	Median	Std dev	Min	Max
Weekly spend (Rs)	402.7	300.0	421.3	0.0	3000.0
Orders per week	2.5	2.0	1.7	0.0	10.0
Orders alone	0.8	1.0	1.2	0.0	7.0
Orders with friends	1.6	1.0	1.1	0.0	7.0
Spend per order (Rs)	168.3	150.0	92.4	10.0	500.0
Meals skipped per week	4.3	3.0	3.4	0.0	20.0
Days menu disliked (of 7)	3.6	3.0	1.4	1.0	7.0
Taste rating (1-5)	2.20	2.00	0.83	1.00	4.00
Quality rating (1-5)	2.45	2.00	0.81	1.00	4.00
Overall mess index (1-5)	2.65	2.67	0.61	1.33	4.50

Weekly spend is strongly right-skewed: the mean (Rs 403) is well above the median (Rs 300) because a few heavy spenders pull it up, so we quote the **median** as the typical value. A typical student orders 2 times a week, skips 3 mess meals and dislikes the menu on 3 of 7 days. Taste (mean 2.20) and quality (mean 2.45) are both below the mid-point of 3.

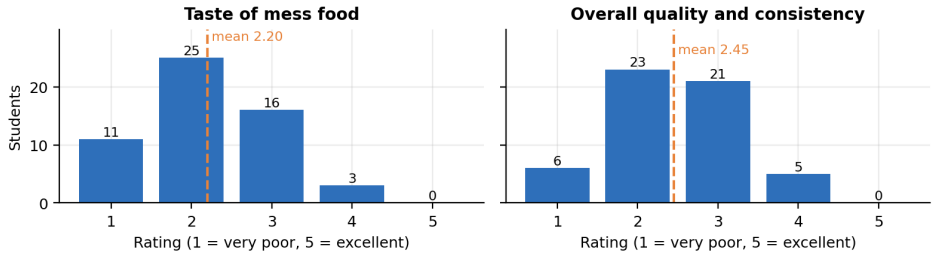
4.2 Mess habits on their own

Figure 2. Mess meals: satisfaction and preference



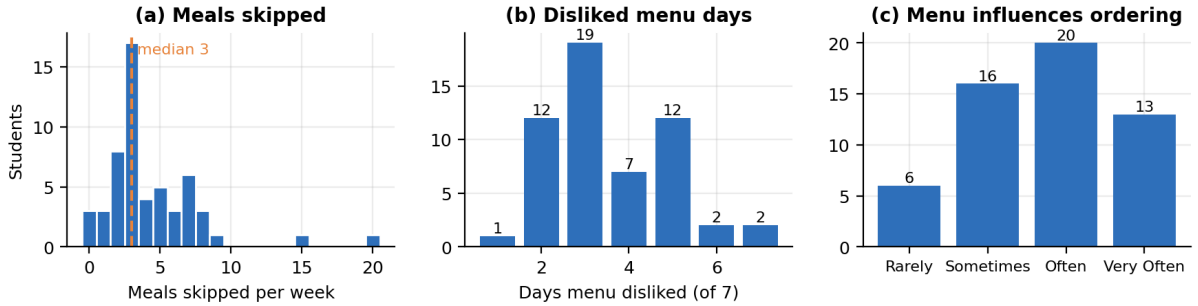
Observation. Satisfaction falls through the day: average rating 3.49 for breakfast, 3.05 for lunch, 2.69 for snacks and 1.98 for dinner. Only 5 (9%) are dissatisfied with breakfast, but 40 (73%) are dissatisfied with dinner and just 3 are satisfied. Breakfast is the favourite meal of 36 (65%) students, lunch of 15, dinner of 4, and snacks of none.

Figure 3. Taste and quality ratings



Observation. 36 (65%) students rate taste 1 or 2 and 29 (53%) rate quality and consistency 1 or 2. Nobody gave the top rating of 5. The two ratings move together (rank correlation 0.59).

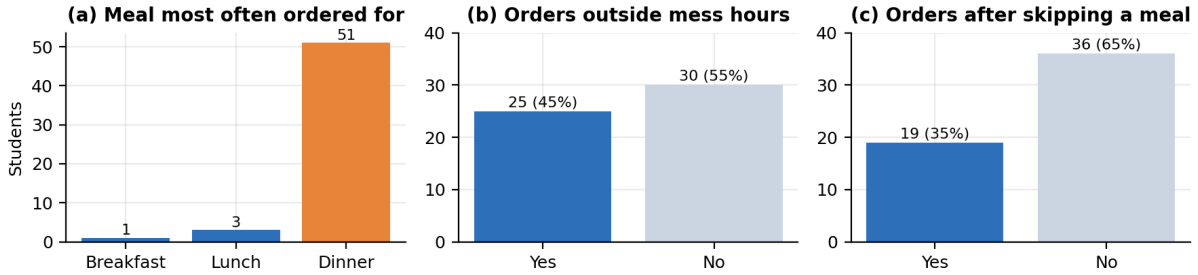
Figure 4. Mess habits



Observation. Skipping meals is common: the median is 3 meals a week, 15 (27%) skip six or more and only 3 skip none. The menu is disliked on 3.6 days a week on average, and 16 (29%) dislike it on five or more days. 33 (60%) say the daily menu influences their decision to order Often or Very Often.

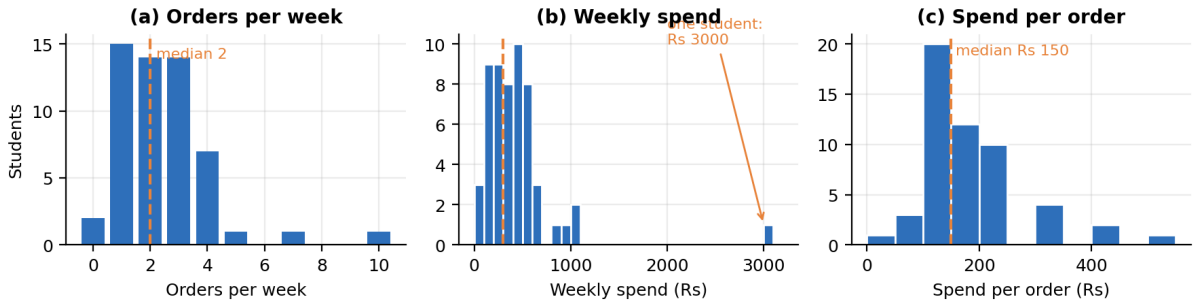
4.3 Outside ordering habits on their own

Figure 5. When and why-context of ordering



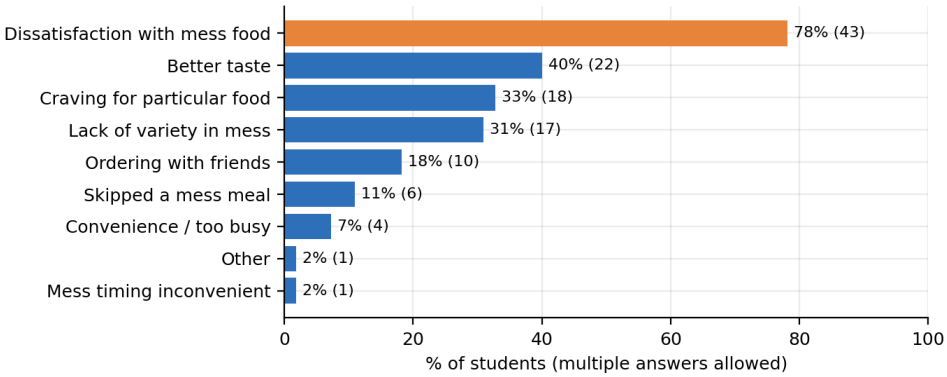
Observation. Ordering is overwhelmingly a dinner habit: 51 (93%) order mostly for dinner, against 3 for lunch and 1 for breakfast. 25 (45%) usually order outside mess timings and 19 (35%) order because they skipped a mess meal.

Figure 6. Ordering frequency and spending



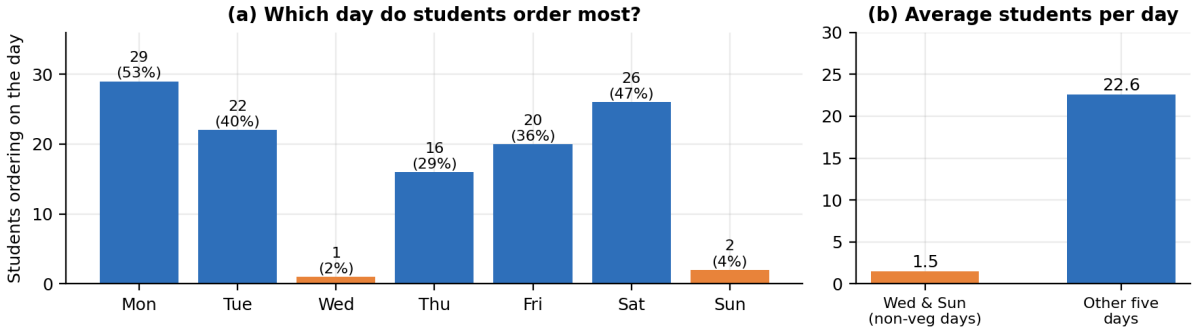
Observation. Orders per week: 2 students order none, 15 order once, 14 twice, 14 three times and 10 four or more times. The typical student spends Rs 300 a week (about Rs 1200 over four weeks), or Rs 150 per order. 16 (29%) spend Rs 500 or more, and the top six spenders account for 33% of the Rs 22150 spent by the whole group in a week.

Figure 7. Reasons for ordering outside food



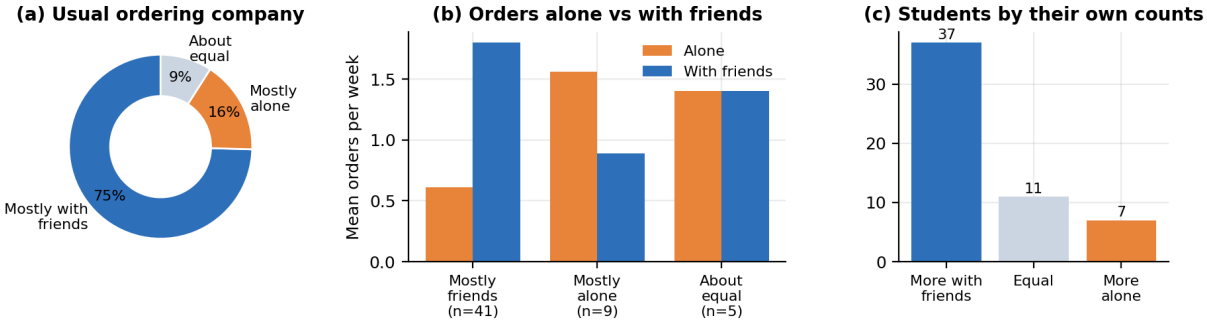
Observation. Dissatisfaction with mess food is by far the most common reason (43 (78%)), followed by better taste (40%), cravings (33%) and lack of variety (31%). Practical reasons such as convenience (4) or mess timing (1) are rare.

Figure 8. Ordering by day of the week



Observation. Monday (29) and Saturday (26) are the most popular ordering days, followed by Tuesday (22) and Friday (20). Wednesday (1) and Sunday (2), the days the mess serves non-veg meals, account for only 3 of 116 ordering-day mentions (2.6%), an average of 1.5 students per day against 22.6 on the other five days. This is consistent with students not needing outside food when the mess serves something they like, although the survey cannot prove the reason.

Figure 9. Social influence on ordering



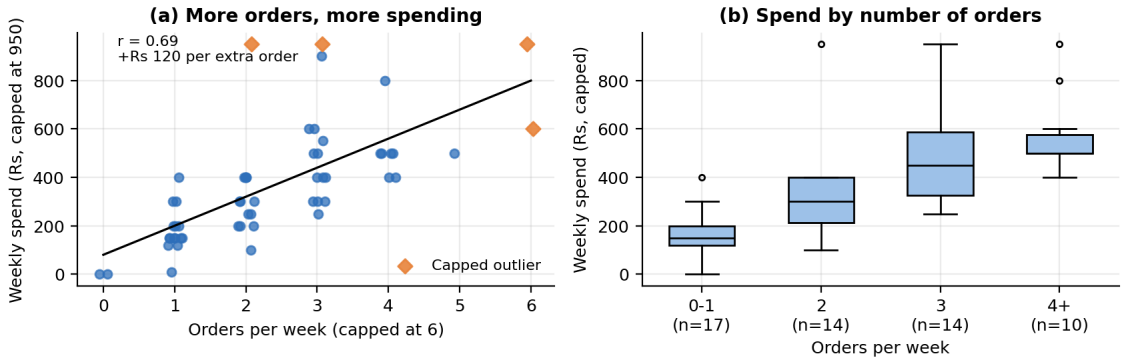
Observation. 41 (75%) students usually order with friends, 9 (16%) alone and 5 (9%) about equally. Of the 135 orders per week in the group, 89 (66%) are with friends. 37 students order more with friends than alone, 11 equally and 7 more alone; 25 (45%) never order alone. Panel (b) confirms that the stated company matches the actual counts.

4.4 Relationships between mess habits and ordering habits

Strength of association is described with the rule of thumb: correlation of 0.6 or more is strong, 0.3 to 0.6 moderate, below 0.3 weak. Correlation shows that two things move together, not that one causes the other.

(a) Number of orders and spending

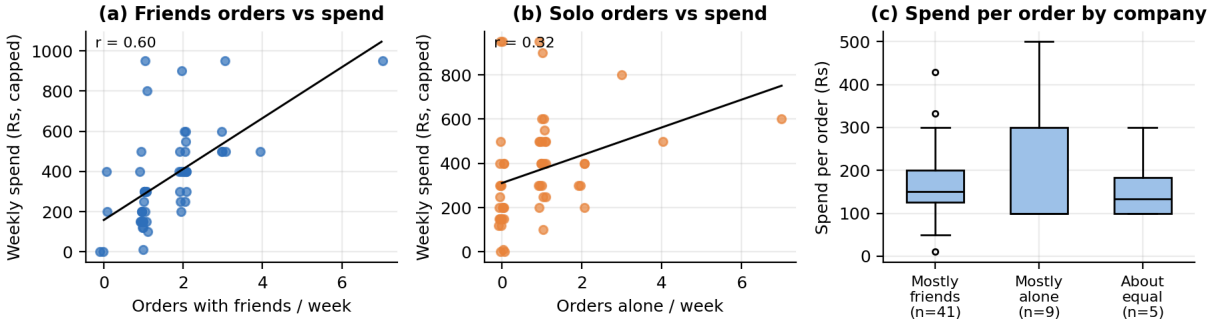
Figure 10. Number of orders and spending



Observation. More orders mean more spending. Median weekly spend is Rs 150 for 0-1 orders, Rs 300 for two, Rs 450 for three and Rs 500 for four or more (mean Rs 770, lifted by heavy spenders). The correlation is 0.69 (strong), with roughly Rs 120 extra spend per additional weekly order, in line with the typical Rs 150 per order.

(b) Social influence and spending

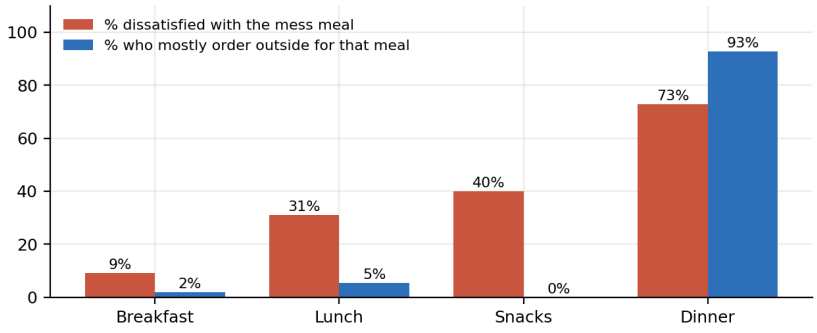
Figure 11. Social context and spending



Observation. Orders with friends are more closely tied to spending (correlation 0.60) than solo orders (0.32). However, the company a student usually keeps does not change how much they order: average orders per week are 2.4 (mostly friends), 2.4 (mostly alone) and 2.8 (equal). Spend per order averages Rs 161 for the friends group and Rs 214 for the solo group, but the solo and equal groups are small (n = 9 and 5), so this is only indicative.

(c) The least-liked meal is the meal students replace

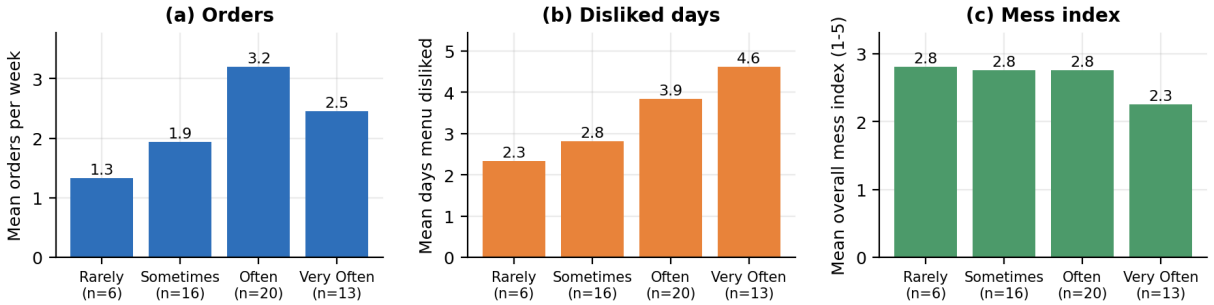
Figure 12. The least-liked meal is the one students replace



Observation. Dinner is where mess dissatisfaction (73%) and outside ordering (93%) meet. 37 of the 40 students who are dissatisfied with dinner mostly order outside for dinner. Breakfast, the best-rated meal, is almost never replaced.

(d) Menu influence

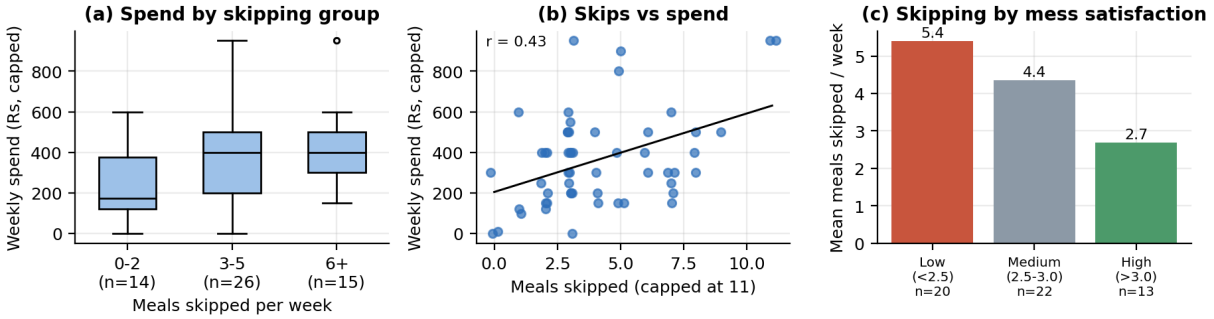
Figure 13. Menu influence on ordering (x-axis) vs orders, disliked days and mess rating



Observation. Students who say the menu influences their ordering more often order more: 1.3 orders a week for 'Rarely', 1.9 for 'Sometimes' and 3.2 for 'Often'. The pattern is not perfectly steady, since the 'Very Often' group averages 2.5. The number of disliked days rises clearly with menu influence (2.3 to 4.6), and the 'Very Often' group also has the lowest overall mess rating (2.3 against about 2.8 for the others).

(e) Meal skipping

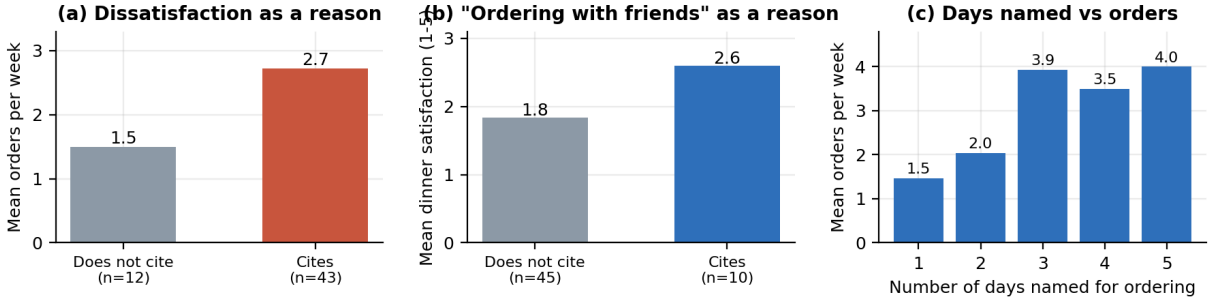
Figure 14. Meal skipping, spending and mess satisfaction



Observation. Students who skip more meals spend more. Median weekly spend is Rs 175 for those skipping 0-2 meals, Rs 400 for 3-5 and Rs 400 for 6 or more (means Rs 229, Rs 394 and Rs 580); the correlation is 0.43 (moderate). Skipping falls as mess satisfaction rises: students with low overall ratings skip 5.4 meals a week, against 2.7 for those with high ratings. An unexpected pattern: students who usually order outside mess hours skip *fewer* meals (3.6) than those who do not (5.0), so skipped meals are not always replaced by an order.

(f) Stated reasons and behaviour

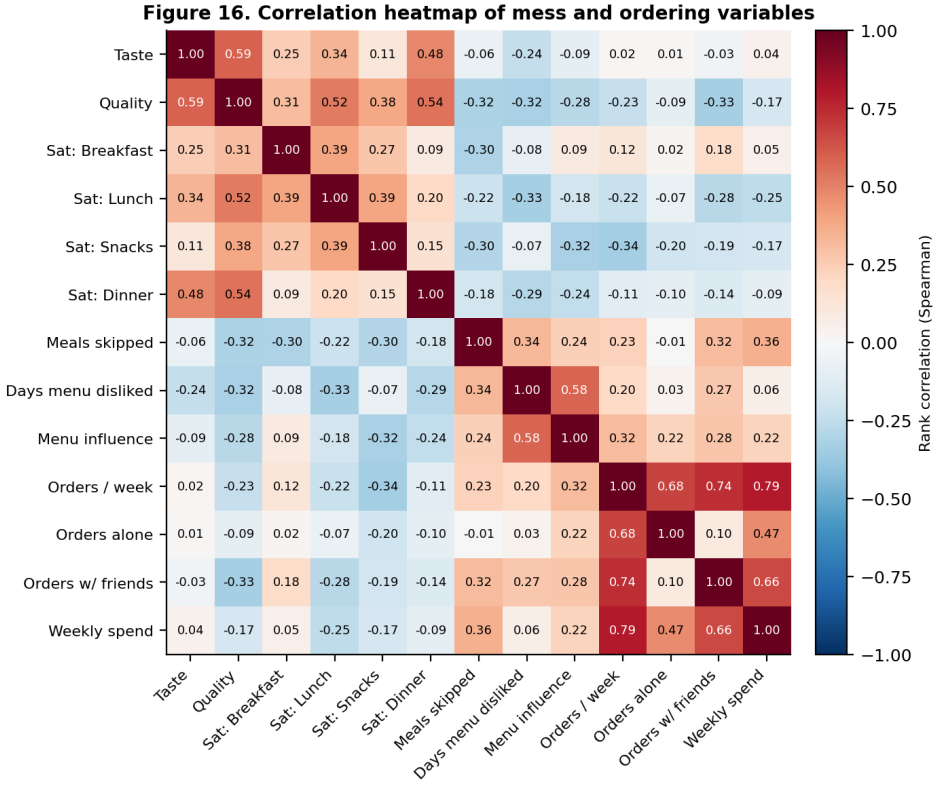
Figure 15. Stated reasons and ordering behaviour



Observation. Students who give dissatisfaction as a reason order more (2.7 a week) than those who do not (1.5). Students who order with friends as a reason rate the mess dinner higher (2.6 against 1.8), which suggests that social ordering is driven by company rather than by unhappiness with the mess. Panel (c) is a consistency check: students who name more ordering days order more times per week.

(g) The whole picture: correlation heatmap and behaviour profile

Figure 16. Correlation heatmap of mess and ordering variables



Strongest links. Orders per week and weekly spend (0.79); orders per week with orders with friends (0.74) and with orders alone (0.68); taste and quality (0.59); disliked days and menu influence (0.58); quality and dinner satisfaction (0.54). **Mess side:** better ratings of breakfast, snacks and dinner go with fewer skipped meals (about -0.30 to -0.18). **Weak links:** taste and orders per week (0.02) and dinner satisfaction and orders (-0.11) are close to zero. In other words, the mess ratings explain *whether* a student is unhappy and skips meals better than they explain *how many* orders the student places.

Figure 17. Behaviour profile by overall mess satisfaction



Observation. On the common 0-1 scale, students with low overall mess satisfaction skip more meals (0.45 vs 0.24), dislike the menu on more days (0.53 vs 0.32), order slightly more (0.44 vs 0.33) and spend slightly more (0.41 vs 0.34) than students with high satisfaction. The gap is large for habits inside the mess (skipping, disliked days) and small for ordering and spending.

5. Conclusion

5.1 Key findings

#	Finding	Evidence
1	Dinner is the weak spot of the mess.	73% dissatisfied with dinner versus 9% with breakfast (Fig. 2).
2	Outside ordering is a dinner habit.	93% mostly order for dinner; 37 of 40 dinner-dissatisfied students do so (Figs. 5, 12).
3	Spending follows the number of orders.	Median spend Rs 150 to Rs 500 from 0-1 to 4+ orders; about Rs 150 per order (Figs. 6, 10).
4	Ordering is social.	66% of orders are with friends; 25 (45%) never order alone (Fig. 9).
5	Wednesday and Sunday see almost no orders.	3 of 116 ordering-day mentions; both are non-veg days (Fig. 8).
6	Dissatisfaction is the main stated reason.	78% cite it, and they order more (2.7 vs 1.5 a week) (Figs. 7, 15).
7	Skipping and a disliked menu go with more ordering and spending.	Median spend Rs 175 vs Rs 400 (0-2 vs 6+ skips); orders rise with menu influence (Figs. 13, 14).
8	Mess ratings relate more to skipping than to ordering.	Satisfaction and skipping are moderately linked; satisfaction and order counts only weakly (Figs. 16, 17).

5.2 What this means

If mess management wants to reduce outside ordering, the data points first to **dinner**: improving its taste, quality and variety targets the meal that students are most unhappy with and most often replace. The low ordering on non-veg days suggests that menu appeal matters, and that a more appealing menu on other days could shift some of the ordering back to the mess. Because ordering is mostly done in groups, improvement may need to be visible to the whole friend circle to change habits.

5.3 Limitations

- Small, non-random sample (55 students, convenience sampling, 84% third-year, one mess); results should not be generalised to all students.
- Spending, skipping and ordering are self-reported for a typical week and may be inaccurate.
- Some groups are small (for example only 9 students order mostly alone), so differences between them are indicative only.
- The Rs 3000 weekly spend and a few very high skip and order counts were capped rather than removed; this reduces their influence but is a judgement call.
- Correlations do not prove causation. The explanation for the low Wednesday and Sunday ordering is not tested by the survey.
- The survey did not record which days the menu was disliked or which meals were skipped, so day-level and meal-level links could not be tested.

5.4 Possible extensions

A larger and more random sample, questions on the specific days and meals disliked, and the actual mess fees, would allow a direct comparison of what students pay the mess and what they pay outside.

Appendix: Questionnaire

Questions used in the analysis (the original Google Form contained additional questions that were removed during cleaning).

#	Question	Answer format
1	How do you usually place your outside food orders?	Mostly alone / About equally alone and with friends / Mostly with friends
2	What is the main reason you order food from outside?	Multiple choice, select all: Dissatisfaction with mess food; Better taste; Craving for a particular food; Lack of variety in the mess; Ordering with friends; I skipped a mess meal; Convenience / too busy; Mess timing is inconvenient; Other
3	When do you most often order outside food?	Breakfast / Lunch / Dinner
4	On which day do you most often order outside food?	Multiple choice, select all: Monday to Sunday

#	Question	Answer format
5	Approximately how much do you spend on outside food orders in a typical week?	Number (Rs)
6	Do you usually order outside food outside the operating timings of the mess?	Yes / No
7	Do you usually order outside food because you skipped a mess meal?	Yes / No
8	How satisfied are you with the taste of your mess food?	Rating 1 to 5
9	How often does the day's mess menu influence your decision to order outside food?	Rarely / Sometimes / Often / Very Often
10	How satisfied are you with each mess meal? (Breakfast, Lunch, Snacks, Dinner)	Very dissatisfied / Dissatisfied / Neutral / Satisfied / Very satisfied, for each meal
11	On average, how many mess meals do you skip in a typical week?	Number
12	Which mess meal do you prefer or like the most?	Breakfast / Lunch / Snacks / Dinner
13	How would you rate the overall quality and consistency of your mess food?	Rating 1 (very poor) to 5 (excellent / very consistent)
14	How many times do you order food from outside in a typical week?	Number
15	In a typical week, how many times do you order outside food alone?	Number
16	In a typical week, how many times do you order outside food with friends?	Number
17	In a typical week, on how many days do you dislike the mess menu or find it unappealing?	Number of days
18	What is your current year of study?	2nd year / 3rd year / 4th year / Other